

Reed J. Kenny

Curriculum Vitae

Department of Biological Sciences, California Polytechnic State University
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EDUCATION

- Ph.D., Ecology** 2025
University of California, Davis — Ecology Graduate Group (2020–2025)
Dissertation: *Phylogenetic systematics of Juncaceae and genomic applications in assigning morphologically intermediate individuals to species.*
Advisor: Dr. Dan Potter
- M.S., Biology** 2020
California Polytechnic State University, San Luis Obispo (2018–2020)
Thesis: *A Floristic Study of the Cal Poly Swanton Pacific Ranch, and a New Combination in Sanicula crassicaulis.*
Advisors: Dr. Matt Ritter, Dr. Jenn Yost
- B.S., Plant Sciences** 2017
University of California, Santa Cruz (2013–2017)
Magna cum laude; Highest Honors in the Major
Research advisor: Dr. Michael Loik

ACADEMIC AND PROFESSIONAL APPOINTMENTS

- Director, Halter Ranch Conservation Scholars (HARCS) Program** Dec. 2025–present
Department of Biological Sciences, Cal Poly, San Luis Obispo
 - Designed and now direct a conservation training program funded by a \$200,000+ industry gift, pairing a quarter of Cal Poly coursework with a paid summer internship at a partner conservation organization.
 - Recruited and manage a partner network spanning land trusts, tribal co-managers, state agencies, and private landowners; negotiate project scopes and student placements.
 - Lead the associated seminar and mentor scholars on independent field projects; first cohort (seven students, 2026) completed projects on beaver dam analog restoration, virtual fencing, rangeland monitoring, fuels treatment, and wildlife inventory.
 - Manage program budget, donor relations, and expansion, including active discussions with additional agricultural partners and an individual-giving channel.
- Lecturer, Department of Biological Sciences** Mar. 2025–present
Cal Poly, San Luis Obispo
 - Instructor of record for BIO 1114 Plant Diversity and Ecology (lecture and lab) and BIO 4472 Conservation Ecology Seminar, which I developed for the HARCS program.
- Postdoctoral Researcher, Urban Forest Institute (UFI)** Jan. 2025–present
Cal Poly, San Luis Obispo — Supervisor: Matt Ritter
 - Landscape-scale analysis of urban forest structure and wildfire impacts, including a study of urban tree and structure loss across the 2025 Eaton and Palisades fires (published 2026).
 - Develop statewide spatial data products and public-facing tools; present results at professional and practitioner conferences.

Botanist (GS-7), Tahoe National Forest

Apr.–Sept. 2018

USDA Forest Service, Nevada City, California

- Conducted floristic surveys on National Forest System lands; identified and mapped rare plant populations in ArcGIS Collector and documented occurrences for NEPA analysis.

Vegetation Crew Lead

Apr.–Aug. 2017

Eastern Nevada Landscape Coalition, Ely, Nevada

- Led a two-person crew conducting post-fire vegetation surveys across Great Basin rangelands, including plant identification and soil classification.

Student Assistant, UC Santa Cruz Arboretum

May 2014–Mar. 2017

University of California, Santa Cruz

- Managed the maritime chaparral collection; supervised interns and student workers; conducted wild seed and cutting collection, seed banking, propagation, outplanting, and invasive plant control.

Botanist (GS-5), Plumas National Forest

May–Sept. 2015

USDA Forest Service, Nevada City, California

- Conducted floristic surveys and rare plant mapping with Trimble GPS; supported GIS and data management.

TEACHING**Cal Poly, San Luis Obispo**

<i>Instructor of record</i> - Plant Diversity and Ecology, 120 students	Fall 2026
<i>Instructor of record</i> - Conservation Ecology and Practice Seminar, 7 students.	Spring 2026
<i>Teaching Assistant</i> - Introductory Botany Lab, 24 students	Fall 2018, 2019
<i>Teaching Assistant</i> - Plant Diversity and Ecology Lab, 20 students	Winter/Spring 2019, 2020

UC Davis

<i>Teaching Assistant</i> - Angiosperm Systematics, 30 students	Spring 2024
<i>Guest Lecturer</i> - Angiosperm Systematics	Spring 2024
<i>Teaching Assistant</i> - California Floristics, 30 students	Spring 2021, 2022, 2023
<i>Teaching Assistant</i> - Evolution of the Human Life Cycle, 20 students	Fall 2023
<i>Teaching Assistant</i> - Environmental Plants, 20 students	Fall 2021

UC Santa Cruz

<i>Teaching Assistant</i> - Systematic Botany of Flowering Plants Lab, 25 students	Winter 2017
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PEER-REVIEWED PUBLICATIONS

*Author name in bold; *denotes undergraduate co-author I mentored.*

Kenny, R., & Potter, D. (in press). Gene flow blurs species boundaries in the *Juncus ensifolius*–*saximontanus* (Juncaceae) species complex. *Systematic Botany*.

- Kenny, R.**, Johns, J., Pawlak, C., Fricker, A., Yost, J. M., & Ritter, M. (2026). Urban trees and structure loss in the 2025 Eaton and Palisades fires. *Urban Forestry & Urban Greening*. <https://doi.org/10.1016/j.ufug.2026.129470>
- Kenny, R.**, Závěská Drábková, L., & Potter, D. (2026). Phylogenetic systematics of Juncaceae. *American Journal of Botany*. <https://doi.org/10.1002/ajb2.70166>
- McMahan, W., Escalona, M., **Kenny, R.**, Marimuthu, P., Nguyen, O., Fairbairn, C., Seligmann, W., Miller, C., Shaffer, H. B., Still, S., & Potter, D. (2025). A genome assembly of the California flannelbush, *Fremontodendron californicum*. *Journal of Heredity*. <https://doi.org/10.1093/jhered/esaf105>
- Kenny, R.**, McMahan, W., Ayres, D., Meyer, V., Grotkopp, E., Still, S., & Potter, D. (2025). Investigating putative hybrids of *Fremontodendron decumbens* and *F. californicum* in the Sierra foothills. *Madroño*. <https://doi.org/10.3120/0024-9637-250014>
- Kenny, R.**, West, J. A., Neubauer, D., Yost, J. M., Grossenbacher, D., & Ritter, M. (2023). A floristic study of Swanton Pacific Ranch, Santa Cruz County, California. *Madroño*. <https://doi.org/10.3120/0024-9637-69.2.139>
- Kenny, R.**, West, J. A., Johnson, H.*, Yost, J. M., & Ritter, M. (2022). A new combination in *Sanicula crassicaulis* (Apiaceae). *Madroño*. <https://doi.org/10.3120/0024-9637-69.1.102>

OTHER PUBLICATIONS AND TECHNICAL PRODUCTS

- Kenny, R.** (2020). Book review: *Plants of Northern California: A Field Guide to Plants West of the Sierra Nevada*. *Madroño*. <https://doi.org/10.3120/0024-9637-67.1.61>
- Kenny, R.** (2023). *An introduction to mapping in R, continued*. Davis R Users Group tutorial. d-rug.github.io/Intro_to_mapping_2/. Spatial data handling and cartography in R.
- Kenny, R.** (2023). *RStudio on FARM*. Davis R Users Group tutorial. d-rug.github.io/R-Studio-on-Farm/. Using RStudio on the UC Davis high-performance computing cluster.

INVITED AND OUTREACH PRESENTATIONS

- Kenny, R.** (2025). What a Rush: adventures in the systematics of Juncaceae. California Native Plant Society, San Luis Obispo Chapter, San Luis Obispo, CA.
- Kenny, R.** (2022). Systematics of Juncaceae. California Native Plant Society, Santa Clara Valley Chapter, virtual.
- Kenny, R.**, Ritter, M., & Yost, J. M. (2019). A floristic study of Halter Ranch. California Native Plant Society, San Luis Obispo Chapter, San Luis Obispo, CA.

CONTRIBUTED PRESENTATIONS AND POSTERS

- Kenny, R.**, Yost, J. M., & Ritter, M. (2025). Online tools for every arborist and urban forester. Partners in Urban Forestry, Henderson, NV. (talk)
- Kenny, R.**, & Potter, D. (2024). Gene flow blurs species boundaries in the *Juncus ensifolius*–*saximontanus* (Juncaceae) species complex. California Botanical Society Graduate Research Symposium, San Luis Obispo, CA. (talk)
- Kenny, R.**, McMahan, W., Ayres, D., Meyer, V., Grotkopp, E., Still, S., & Potter, D. (2024). Investigating putative hybrids of *Fremontodendron decumbens* and *F. californicum* in the Sierra foothills. Northern California Botanists Symposium, Chico, CA. (talk)
- Kenny, R.**, Závěská Drábková, L., & Potter, D. (2024). Systematics of Juncaceae: phylogenetic relationships inferred from Hyb-Seq data. International Botanical Congress, Madrid, Spain. (poster)

- Kenny, R.** (2022). Phylogenetics of Juncaceae. California Native Plant Society Conservation Conference, San Jose, CA. (talk)
- Hazelquist, C., Williams, C. M., Hasenstab-Lehman, K., **Kenny, R.**, & Boes, W. (2021). Using high-throughput DNA sequence data to infer the phylogeny of *Lewisia* (Montiaceae) and evaluate a potentially new species from the Sierra Nevada foothills. Southern California Botanists Symposium, virtual. (poster)
- Kenny, R.**, Ritter, M., & Yost, J. M. (2019). Traversing Swanton: a multigenerational floristic study of the Cal Poly Swanton Pacific Ranch. California Botanical Society Student Symposium, San Luis Obispo, CA. (talk)
- Kenny, R.**, & Loik, M. (2017). Summer root and soil water distribution under annual grasses and *Stipa pulchra*. Northern California Botanists Symposium, Chico, CA. (poster)

UNDERGRADUATE RESEARCH

Summer 2026 Rowan Campbell (ENVM)

- Cascades frog surveys in the Klamath Mountains in conjunction with CDFW
- PIT-tagged frogs and collected health data
- Removed invasive trout
- Shared data collected with Survey123

Summer 2026 Rylyn Jaggar (CRP) & Rylie Helburg (ENVM)

- Impacts of fuels treatments on biodiversity at Bearpaw Preserve
- Ran network of 10 camera traps and managed data
- Conducted plant quadrat and line-intercept surveys
- Conducted bird point-count surveys

Summer 2026 Landon Miller (BIO)

- Beaver dam analog meadow restoration effects on mammal biodiversity with the Scott River Watershed Council
- Collected camera trap data before and after meadow restoration
- Analyzed camera data

Summer 2026 Santiago Olvera (EESS)

- Baseline report for a new conservation property with the Land Conservancy of SLO County
- Collected photo and GIS data on roads, structures and vegetation
- Developed GIS maps
- Compiled data into a technical report

Summer 2026 Finn Torres (BIO) & Natalie Donnellson (ENVM)

- Rangeland monitoring for long-term analysis of virtual fencing impacts at Rana Creek Preserve
- Collected baseline rangeland samples for fuel analyses
- GIS analyses and field data collection for siting beaver dam analog restoration

Summer 2019 Thomas Buchloh (BIO), Sam Farrow (BIO) & Celeste Dorion (BIO)

- Swanton Ranch Summer Internship: Measuring and monitoring the vegetation of the Scott Creek Marsh
- Lived at Swanton for the summer
- Conducted vegetation transects to create GIS map of all marsh plants ahead of Caltrans creek restoration under HWY 1
- Used drones to conduct spectral analyses

- Presented work at the Northern California Botanists Symposium, Chico, CA

Summer 2019 Hunter Johnson (BIO)

- An investigation of speciation at Swanton Pacific Ranch in *Sanicula crassicaulis*
- Collected plant morphometric data
- Co-authored the resulting publication (*Madroño*, 2022)

EXTERNAL FUNDING GRANTS

Research and program funding secured: **\$263,486**

Program and Industry Funding

Halter Ranch Conservation Scholars Program Donation \$252,631 2025

Gift from Halter Ranch Foundation to Cal Poly, funds program operations, student stipends, and the program director position for two years. Support likely to be ongoing.

Competitive Research Grants

UC Davis Jastro-Shields Research Award (\$1,995)	2024
UC Davis Jastro-Shields Research Award (\$3,000)	2022
California Native Plant Society, Santa Clara Valley Chapter Research Award (\$1,500)	2022
Davis Botanical Society Research Grant (\$2,760)	2021
Northern California Botanists Research Grant (\$1,000)	2021
California Native Plant Society Research Grant (\$600)	2021

FELLOWSHIPS, SCHOLARSHIPS, AND AWARDS

Total: **\$134,222**

UC Davis Plant Sciences Graduate Fellowship (\$70,000)	2020
UC Davis Ecology Graduate Group Fellowship (\$50,722)	2020
David and Frieda Wertmann Scholarship, Cal Poly San Luis Obispo (\$11,000)	2019
California Native Plant Society, San Luis Obispo Chapter, McLeod Scholarship (\$500)	2019
Koret Undergraduate Research Scholarship, UC Santa Cruz (\$2,000)	2017
Highest Honors in the Major, UC Santa Cruz Plant Sciences	2017

TECHNICAL AND RESEARCH EXPERTISE

Landscape-Scale Spatial Analysis

R, Python, Google Earth Engine, and ArcGIS Pro pipelines for processing remote sensing and other spatial data at landscape and statewide extents; statistical treatment of spatial autocorrelation and interpolation of discrete data sources.

Biophysical Modeling

Large-scale modeling of biophysical processes, including wind velocity using NOAA's High-Resolution Rapid Refresh atmospheric model with WindNinja, and streamflow using high-resolution digital elevation models and hydrologic models implemented in R.

Biological Inventory and Monitoring

Design and implementation of field data collection and processing for inventory and monitoring: mobile data platforms in Esri Field Maps and Survey123; rare and invasive plant mapping; meadow

and rangeland monitoring design; herbarium voucher collection and curation; iNaturalist projects for crowdsourced data. Camera trapping, and AI-assisted identification (Addax AI, Wildlife Insights).

Data Science and Computing

High-performance computing clusters and parallel processing; analysis of biodiversity datasets of millions of records to produce diversity statistics, range maps, and species interaction networks; reproducible workflows in R and Python, version control with Git.

Genomics and Bioinformatics

Target capture, RAD-seq, and whole-genome resequencing analysis in HPC environments; phylogenomic and population genomic inference, species delimitation, and hybridization analysis.

SERVICE AND OUTREACH

Developed and presented two open-access computational tutorials for the Davis R Users Group (2023).

Recurring speaker for California Native Plant Society chapters (San Luis Obispo 2019; Santa Clara Valley 2022, San Luis Obispo 2025) and practitioner audiences at Partners in Urban Forestry (2025).

Designed and administer the HARCS Symposium, an annual research symposium hosted by an industry partner (Paso Robles, CA).

Reviewer for Madroño (1 manuscript) and Journal of the Botanical Research Institute of Texas (1 manuscript).

REFERENCES

Dr. Dan Potter — Professor, Department of Plant Sciences, University of California, Davis
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Dr. Matt Ritter — Professor, Department of Biological Sciences, Cal Poly San Luis Obispo
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Relationship: M.S. advisor

Frazier Haney — Executive Director, The Wildlands Conservancy
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